Mathematical Solution: Certifying the n-Qubit GHZ State

1. Target State & Problem Formulation

The n-qubit Greenberger–Horne–Zeilinger (GHZ_n) state is the many-qubit generalization of a Bell pair:

$$|\mathrm{GHZ}_n\rangle = \frac{1}{\sqrt{2}}(|00\ldots0\rangle + |11\ldots1\rangle) = \frac{1}{\sqrt{2}}\left(|0\rangle^{\otimes n} + |1\rangle^{\otimes n}\right)$$

To uniquely certify that a quantum device has prepared $|\mathrm{GHZ}_n\rangle$ using only the expectation values of Pauli observables, we measure n stabilizer generators:

- **Global X-basis parity (1 check):** $\Pi_X = X_0 X_1 \cdots X_{n-1} = \bigotimes_{i=0}^{n-1} X_i$
- **Neighbouring ZZ parities ($n-1$ checks):** $Z_i Z_{i+1}$ for each adjacent pair $i \in \{0, 1, \ldots, n-2\}$

2. Step-by-Step Quantum Circuit Derivation

Step 2.1: Initialization

All n wires begin in the computational basis ground state:

$$|\psi_0\rangle = |0\rangle^{\otimes n} = |0_0 0_1 \ldots 0_{n-1}\rangle$$

Step 2.2: Hadamard Gate on Wire 0

Applying the Hadamard operator H_0 on wire 0 maps $|0_0\rangle \mapsto \frac{1}{\sqrt{2}}(|0_0\rangle + |1_0\rangle)$, creating an equal superposition on the first wire while leaving the remaining $n-1$ wires in $|0\rangle$:

$$|\psi_1\rangle = (H_0 \otimes I_1 \otimes \cdots \otimes I_{n-1})|0_0 0_1 \ldots 0_{n-1}\rangle = \frac{1}{\sqrt{2}}(|0_0 0_1 \ldots 0_{n-1}\rangle + |1_0 0_1 \ldots 0_{n-1}\rangle)$$

Step 2.3: CNOT Fan-Out Entanglement

Next, we apply a controlled-NOT gate $\mathrm{CNOT}_{0,i}$ from control wire 0 to every target wire $i \in \{1, 2, \ldots, n-1\}$. Recall that $\mathrm{CNOT}_{0,i}|c_0, t_i\rangle = |c_0, t_i \oplus c_0\rangle$:

- On the first branch $|0_0 0_1 \ldots 0_{n-1}\rangle$, the control qubit is 0, so all target qubits remain $|0\rangle$.
- On the second branch $|1_0 0_1 \ldots 0_{n-1}\rangle$, the control qubit is 1, flipping every target qubit from $|0\rangle$ to $|1\rangle$.

$$|\psi_{\text{final}}\rangle = \left(\prod_{i=1}^{n-1} \mathrm{CNOT}_{0,i}\right)|\psi_1\rangle = \frac{1}{\sqrt{2}}(|00\ldots0\rangle + |11\ldots1\rangle) = |\mathrm{GHZ}_n\rangle$$

3. Derivation of the n Parity Expectation Values

Step 3.1: The $n-1$ Neighbouring ZZ Parities

The single-qubit Pauli-Z operator acts on computational basis states as $Z|0\rangle = +|0\rangle$ and $Z|1\rangle = -|1\rangle$, or concisely $Z|b\rangle = (-1)^b|b\rangle$ for $b \in \{0,1\}$. For any adjacent pair $(i, i+1)$, the product Z_iZ_{i+1} acts on an n-qubit basis state as:

$$Z_iZ_{i+1}|b_0 \dots b_i b_{i+1} \dots b_{n-1}\rangle = (-1)^{b_i+b_{i+1}}|b_0 \dots b_i b_{i+1} \dots b_{n-1}\rangle$$

Evaluating Z_iZ_{i+1} on both branches of $|\mathrm{GHZ}_n\rangle$:

- **All-zeros branch ($b_i = 0, b_{i+1} = 0$):** $Z_iZ_{i+1}|00\dots0\rangle = (+1)(+1)|00\dots0\rangle = +|00\dots0\rangle$
- **All-ones branch ($b_i = 1, b_{i+1} = 1$):** $Z_iZ_{i+1}|11\dots1\rangle = (-1)(-1)|11\dots1\rangle = +|11\dots1\rangle$

By linearity, $|\mathrm{GHZ}_n\rangle$ is a $+1$ eigenstate of every adjacent Z_iZ_{i+1} operator:

$$Z_iZ_{i+1}|\mathrm{GHZ}_n\rangle = +|\mathrm{GHZ}_n\rangle \implies \langle Z_iZ_{i+1}\rangle = \langle \mathrm{GHZ}_n|Z_iZ_{i+1}|\mathrm{GHZ}_n\rangle = +1 \quad \forall i \in \{0,\dots,n-2\}$$

Role of the ZZ checks: Requiring $\langle Z_iZ_{i+1}\rangle = +1$ for all i enforces $b_0 = b_1 = \cdots = b_{n-1}$, restricting the state's support strictly to $\mathrm{span}\{|00\dots0\rangle, |11\dots1\rangle\}$.

Step 3.2: The Global X Parity

The Pauli-X operator acts as a bit flip: $X|0\rangle = |1\rangle$ and $X|1\rangle = |0\rangle$. Consequently, the global X parity operator $\Pi_X = X_0X_1\cdots X_{n-1}$ flips all n qubits simultaneously:

$$\Pi_X|00\dots0\rangle = |11\dots1\rangle, \qquad \Pi_X|11\dots1\rangle = |00\dots0\rangle$$

Applying Π_X to $|\mathrm{GHZ}_n\rangle$ interchanges the two branches, leaving the state invariant:

$$\Pi_X|\mathrm{GHZ}_n\rangle = \frac{1}{\sqrt{2}}(\Pi_X|00\dots0\rangle + \Pi_X|11\dots1\rangle) = \frac{1}{\sqrt{2}}(|11\dots1\rangle + |00\dots0\rangle) = +|\mathrm{GHZ}_n\rangle$$

Therefore, the expectation value of Π_X is:

$$\langle \Pi_X\rangle = \langle \mathrm{GHZ}_n|\Pi_X|\mathrm{GHZ}_n\rangle = +1$$

Role of the Π_X check: Certifies that $|00\dots0\rangle$ and $|11\dots1\rangle$ are in a coherent superposition with a $+1$ relative phase rather than an incoherent classical mixture ρ_{mix}.

4. Mean Parity Score s_n & Uniqueness Table

Averaging all n parity expectation values gives the certification score s_n:

$$s_n = \frac{1}{n}\left(\langle \Pi_X\rangle + \sum_{i=0}^{n-2}\langle Z_iZ_{i+1}\rangle\right) = \frac{1}{n}\left(1 + \sum_{i=0}^{n-2} 1\right) = \frac{1+(n-1)}{n} = 1.0$$

State	$\langle \Pi_X \rangle$	$\sum_{i=0}^{n-2} \langle Z_i Z_{i+1} \rangle$	Mean Score s_n
$\lvert \mathrm{GHZ}_n \rangle = \frac{1}{\sqrt{2}}(\lvert 0 \ldots 0 \rangle + \lvert 1 \ldots 1 \rangle)$	$+1$	$n-1$	**1.0**
Unentangled product state $\lvert + \rangle^{\otimes n}$	$+1$	0	$1/n$
Classical mixture $\rho_{\mathrm{mix}} = \frac{1}{2}(\lvert 0 \ldots 0 \rangle\langle 0 \ldots 0 \rvert + \lvert 1 \ldots 1 \rangle\langle 1 \ldots 1 \rvert)$	0	$n-1$	$(n-1)/n$
Wrong-phase state $\frac{1}{\sqrt{2}}(\lvert 0 \ldots 0 \rangle - \lvert 1 \ldots 1 \rangle)$	-1	$n-1$	$(n-2)/n$

5. Complete PennyLane Python Solution

```
def prepare_and_build_checks(n):
    # Step 1: Build the GHZ state (Hadamard on wire 0, CNOT fan-out to wires 1..n-1)
    qp.Hadamard(wires=0)
    for i in range(1, n):
        qp.CNOT(wires=[0, i])

    # Step 2: Build the global X parity observable
    checks = [qp.prod(*(qp.PauliX(wires=i) for i in range(n)))]

    # Step 3: Build the n-1 neighbouring ZZ parity observables
    for i in range(n - 1):
        checks.append(qp.prod(qp.PauliZ(wires=i), qp.PauliZ(wires=i + 1)))

    # Step 4: Return all n observables
    return checks
```